PAPER_709: Westerlund 2 Young Super Star Cluster: UQFF MUGE

Class: `Westerlund2StarClusterUQFF` **CP4 Entry:** #293 **Keywords:** Westerlund 2, young star cluster, LMC, rapid SF, OB winds, Ug Lambda, UQFF MUGE **Session:** 175 | **Version:** v5.32 **Source:** grok_share_ba508f76c8e.txt

Abstract

Westerlund 2 (Wd 2) is one of the youngest and most massive star clusters in the Milky Way, located at 15,000 ly in the Carina-WR constellation. Its mass ($\sim 3\times10^4 M_\odot$) and age (~ 2 Myr) make it an ideal test case for rapid gas-to-star conversion dynamics. The UQFF MUGE is derived with the dynamic mass growth function incorporating the Wolf-Rayet star contribution.

1. System Parameters

Parameter	Value	Description
M_{init}	$3 \times 10^4 M_\odot$	Initial stellar mass
r_{wd2}	9.461×10^{16} m	Cluster radius (10 ly)
M_0^{gas}	3.333	Gas-to-star mass ratio
τ_{SF}	6.312×10^{13} s	SF timescale (2 Myr)
B_{wd2}	10^{-5} T	Cluster magnetic field

2. Master UQFF Gravity Equation

$$g_{W2}(t) = \frac{GM(t)}{r^2}(1+H_0 t)\left(1-\frac{B}{B_{crit}}\right)+(U_{g1}+U_{g4})(1+f_{TRZ})+\frac{\Lambda c^2}{3}+q(v\times B)\left(1+\frac{\rho_{UA}}{\rho_{SCm}}\right)\times10^{-}$$

2.1 Rapid Gas Conversion (Wolf-Rayet + OB mass infall)

$$M(t) = M_{init}(1 + 3.333\, e^{-t/\tau_{SF}})$$

At $t = 0$: $M = 4.333\, M_{init}$; at $t = \tau_{SF}$: $M = 2.226\, M_{init}$.

3. Numerical Result

$$g_{W2}(t = 1\ \text{Myr}) \approx 1.053 \times 10^{-3}\ \text{m/s}^2$$

References

- Clarkson et al. (2012), ApJ, 751, 132
- Zeidler et al. (2015), A&A, 576, A28
- UQFF Framework, Star-Magic Session 175

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